

A rapid review of clustering algorithms

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ABSTRACT

Clustering algorithms are essential for organising data into meaningful groups based on inherent patterns and similarities. They play a crucial role across domains such as marketing, e-commerce, healthcare, data analysis, and social media. There is a wide range of clustering algorithms, each with unique strengths and limitations, and no single algorithm is universally optimal for all tasks. In this study, we systematically analyse existing clustering algorithms and classify mainstream approaches across five key dimensions: underlying clustering principles, cluster assignment mechanism, dataset scalability, predefined cluster requirements, and application domains. This classification provides a comprehensive perspective on clustering methodologies, helping researchers select suitable algorithms for specific tasks. In addition, we examine current trends, emerging developments, and future directions in clustering research, highlighting unresolved challenges and open research questions.

1. Introduction

In the era of digital transformation, the Internet has greatly simplified how individuals and businesses access information. A large number of documents, videos, and audio files are generated daily on digital platforms, bringing unprecedented challenges to information retrieval. The problem has changed from information scarcity to managing increasingly complex information. Although our computing capabilities are growing rapidly, the sheer volume of data often makes it difficult to extract relevant information from dispersed sources, hindering efficient information retrieval. An effective approach to this problem is to apply machine learning techniques, particularly clustering algorithms, which can automatically group similar data points without requiring prior labels. These algorithms aim to partition data into subgroups, or clusters, such that items within the same cluster are more similar to each other than to those in other clusters. Clustering methods have played a key role in advancing fields and are widely used in everyday applications, including spam detection, recommendation systems, customer segmentation for targeted marketing, and image retrieval based on visual similarity in photo albums, among others.

Clustering algorithms can process various data types, including text, audio, video, and images. In audio processing, clustering techniques leverage acoustic features to group files, aiding in tasks such as genre

classification and speaker identification. For video data, clustering facilitates content organisation, recommendation systems, and summarisation by identifying visual or thematic similarities. In image analysis, it plays a crucial role in segmentation and content-based retrieval. Additionally, clustering is an effective tool for anomaly detection, helping identify irregularities in network traffic, financial transactions, and medical records. This versatility underscores the importance of clustering in identifying patterns and extracting valuable insights across various domains.

Clustering is typically framed as an unsupervised learning task in which no labels are provided and the goal is to discover structure in the data. In contrast, supervised and semi-supervised methods rely on known labels and are more accurately described as classification.

- **Semi-supervised clustering:** Semi-supervised clustering methods operate in settings where a small amount of labelled data — or prior knowledge in the form of pairwise constraints — is available alongside a large amount of unlabelled data. Unlike semi-supervised classification, the goal here remains clustering, but the limited supervision helps guide the formation of clusters towards more meaningful groupings. Typical forms of supervision include labelled seed points, must-link constraints (samples that

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should be in the same cluster), and cannot-link constraints (samples that should not be grouped). Representative algorithms include Constrained K-Means [1], Semi-Supervised Fuzzy C-Means (SSFCM) [2], Semi-Supervised Spectral Clustering [3], and Semi-Supervised Gaussian Mixture Models (SS-GMMs) [4]. These methods improve clustering performance by incorporating weak supervision to reduce ambiguity in cluster assignment.

- **Unsupervised clustering:** In unsupervised clustering, no labelled data is provided. The algorithm discovers intrinsic structure in the data and groups similar samples purely based on feature similarity, without any prior knowledge of the underlying categories. In most applications, the number of clusters is unknown and must be estimated. A common example is identifying emerging topics in Twitter data by clustering posts based on word usage patterns or semantic similarity; groups of tweets about similar subjects emerge naturally without predefined topic labels [5,6]. Well-established unsupervised clustering algorithms include K-Means [7,8], Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [9], and Fuzzy C-Means (FCM) [10]. Several techniques are commonly used to estimate the appropriate number of clusters, including the elbow method, silhouette score, and gap statistic, as discussed in Section 4.4.

Clustering algorithms have expanded into increasingly diverse application domains, making algorithm selection based on conventional categorisations insufficient to address specific analytical challenges. In this comprehensive review, we systematically analyse mainstream clustering methods from five key perspectives: underlying principles & characteristics, assignment mechanisms (membership type), dataset scalability, predefined cluster requirements, and application domains. Unlike previous reviews that focused on algorithmic taxonomies or specific application contexts, our multidimensional classification framework offers a finer, more practical perspective that connects theoretical foundations with implementation considerations. By examining the interrelationships among these five dimensions, we provide decision support that better reflects the complex trade-offs practitioners face when selecting algorithms for real-world scenarios. Additionally, we explore current trends, emerging developments, and future directions in clustering research, highlighting unresolved challenges and open research questions spanning multiple dimensions.

2. Literature review

Clustering algorithms are widely applied in various fields, including text mining, product recommendation, anomaly detection, and image processing. Over the years, a diverse range of clustering methods has been developed, each with distinct assumptions, advantages, and limitations. There has been extensive discussion in the research community about these algorithms, including critical analysis of their computational efficiency, clustering quality, and applicability across different fields.

Rui Xu and Wunsch, D. [11] examined clustering techniques applied to datasets in computer science, machine learning, and statistics. They demonstrated the use of these algorithms on several benchmark datasets, including the travelling salesman problem and bioinformatics, an emerging field receiving significant attention. Additionally, they discussed key topics, including cluster validation, similarity measures, and related methodological aspects. Xu and Tian [12] conducted a comprehensive survey of clustering algorithms in 2015, covering a diverse range of techniques. The authors classified these algorithms according to their underlying principles, characteristics, and applications. The survey includes a detailed, comprehensive comparison of all the algorithms discussed. Ezugwu et al. [13] provided an up-to-date, methodical, and thorough analysis of both conventional and cutting-edge clustering algorithms across various domains from a more practical perspective. Covered the application of clustering to various

fields, such as big data, artificial intelligence, and robotics. The review also focused on the remarkable role that clustering plays in a variety of disciplines, including education, marketing, medicine, biology, and bioinformatics. The three highly cited works published in 2005, 2015, and 2022 are among the most comprehensive studies on clustering algorithms since 2000, making significant contributions to the field.

In addition to these seminal works, other notable works mainly focus on specific classifications or applications within clustering. Bora et al. [14] conducted a comparative study between the fuzzy and hard clustering algorithms. The focus is on assessing and contrasting the performance of these clustering algorithms, providing insights into their strengths and limitations. Sisodia et al. [15] explored diverse clustering algorithms in the field of data mining, emphasising fundamental aspects such as the basics, requirements, classification, challenges, and application domains of these algorithms. Rodriguez et al. [16] provide a comprehensive comparative analysis of nine well-known clustering algorithms. The authors evaluated the performance and characteristics of these algorithms through a systematic evaluation, offering insights into their strengths and limitations. This study helps to understand clustering methods and facilitates informed decisions based on specific application requirements.

While prior surveys (e.g., [12,13]) provide comprehensive overviews of clustering algorithms from algorithmic and application-oriented perspectives, they primarily adopt broad taxonomic or domain-driven classifications. In contrast, this rapid review proposes a multidimensional classification framework that integrates five inter-related criteria—underlying principles, assignment mechanisms, scalability, predefined cluster requirements, and application domains. This structure offers a more decision-oriented perspective, linking theoretical foundations with practical implementation constraints, and incorporates recent developments such as deep representation-based clustering within a unified taxonomy.

3. Methodology

In this section, we describe the process of selecting publications on clustering algorithms from various platforms for our review. The whole process is depicted in Fig. 1. To begin, we defined a set of relevant keywords, including “clustering”, “clustering algorithm”, “clustering method”, “consensus clustering”, and “clustering technique”. Using this keyword list, we searched three reputable academic databases: Google Scholar, arXiv, and Scopus. We applied Boolean operators (AND, OR) and truncation techniques to refine our search queries. We included Google Scholar in our selection because it indexes formally published papers and preprints. Notably, some early preprints have made significant contributions to their respective research fields and have garnered substantial citations. However, due to lengthy journal review processes or delayed official publication, these papers may remain in preprint status for an extended period. Nonetheless, including such preprints is essential, as they represent impactful and timely research.

To ensure the relevance and quality of the publications, we applied the following filtering criteria:

- Written in English
- Published within the last ten years
- Focused on novel clustering techniques

We then removed duplicate papers and articles that were not directly related to clustering algorithms. The remaining papers primarily included algorithm introductions, technical improvements, and application studies. Next, we manually screened titles, abstracts, and keywords to further refine our selection. For the final set of publications, we analysed the full text to extract key information, including the fundamental principles of the clustering algorithms, their applications, experimental procedures, and major findings. To assess the reliability of the results, we considered factors such as experimental design, sample size, and statistical methodologies. The results of this literature review, presented in the next section, synthesise emerging patterns and trends in clustering research.

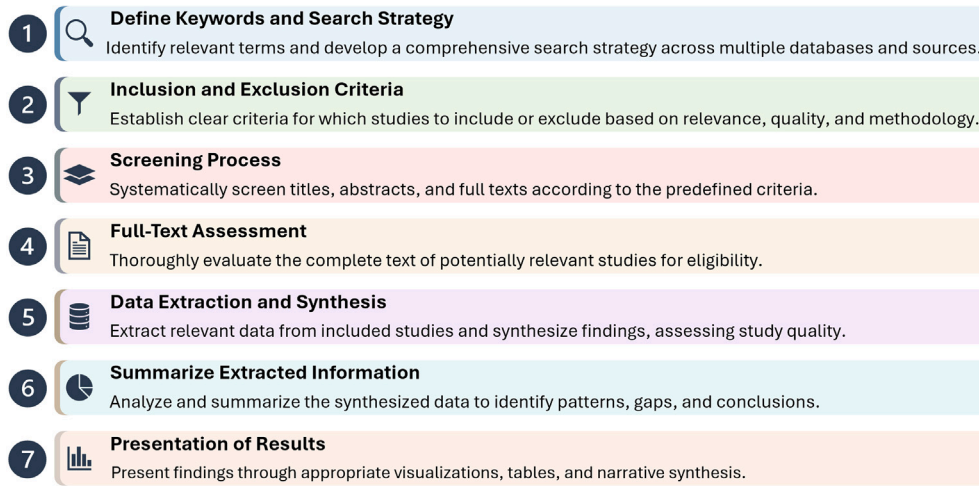


Fig. 1. Literature selection process.

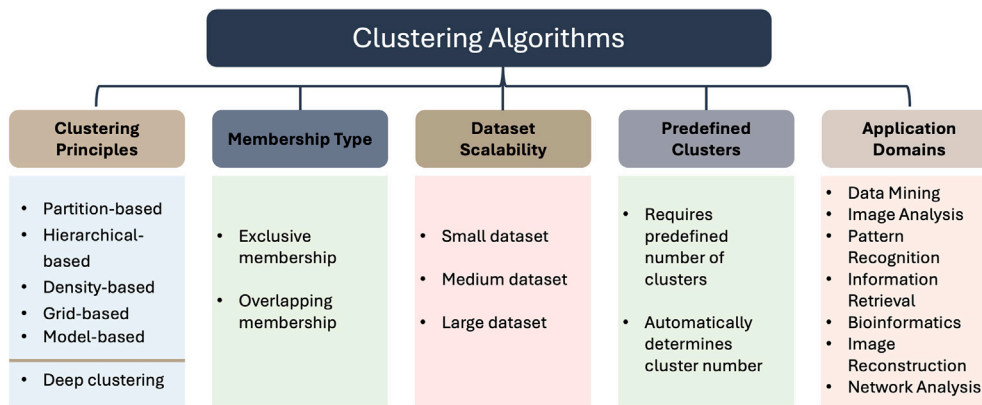


Fig. 2. Structure of the clustering algorithm classification, covering five dimensions.

4. Clustering algorithm classification criteria

To provide a systematic frame of reference, we classify clustering algorithms along five interrelated dimensions (Fig. 2):

- Algorithms can be categorised as partitioning, hierarchical, density-based, grid-based, model-based, or deep representation-based approaches.
- Cluster assignment mechanisms: Algorithms may assign data points with hard membership or soft membership.
- Scalability: This dimension reflects whether the algorithm can handle small, medium, or very large datasets.
- Predefined K requirement: Some algorithms require the number of clusters to be specified in advance, while others do not.
- Application areas: Each algorithm is typically used in specific domains, such as marketing, healthcare, or social media.

We first review the six basic principles and illustrate each underlying clustering principle using simplified, artificially generated data. These visualisations are designed to highlight the core ideas behind each algorithm; however, we acknowledge that real-world clustering scenarios are often significantly more complex and less clean than these illustrative examples. Then we discuss how the remaining four criteria affect the algorithm's practical applicability to a specific task. Together, these dimensions form a comprehensive taxonomy that helps practitioners choose algorithms that meet the constraints and goals of their clustering problems.

4.1. Underlying principles and characteristics

Clustering algorithms can be categorised according to their underlying principles and characteristics, which describe how data samples are grouped and represented. Traditional studies commonly classify clustering methods into several classical paradigms based on their fundamental grouping mechanisms [17–19]. In addition to these classical approaches, recent advances in representation learning have introduced deep clustering [20,21] methods, which integrate neural feature learning with clustering objectives. The overall taxonomy considered in this work is summarised in Table 1.

4.1.1. Classical clustering paradigms

Classical clustering paradigms define clusters based on explicit grouping principles, such as distance similarity, hierarchical structure, data density, spatial partitioning, or probabilistic modelling. These methods form the foundation of clustering research and remain widely used due to their interpretability and strong theoretical basis. In this survey, classical clustering approaches are organised into five main categories, which are introduced in the following sections.

- Partition-Based Clustering (e.g., Fig. 3): Partition-based clustering algorithms partition data into non-overlapping clusters. The basic idea of these clustering algorithms is to consider the centre of data points as the centre of the corresponding cluster, which generally has low time complexity and high computing efficiency. However, they are not well-suited to non-convex data, are sensitive to outliers, can be easily drawn into local optima,

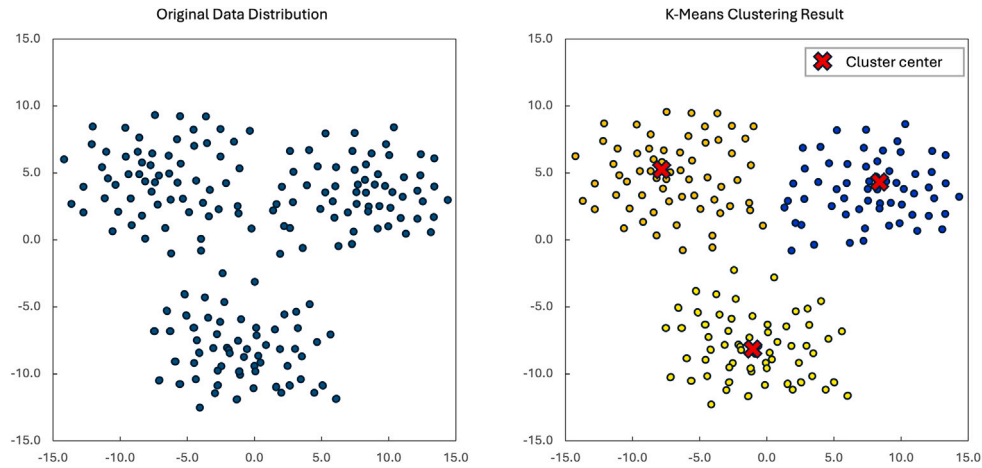


Fig. 3. Illustration of partition-based clustering (K-Means). The left panel shows the original data distribution. The right panel presents the clustering result after assigning each data point to its nearest centroid. Each sample is assigned to its nearest centroid, illustrating the hard-assignment property of partition-based methods.

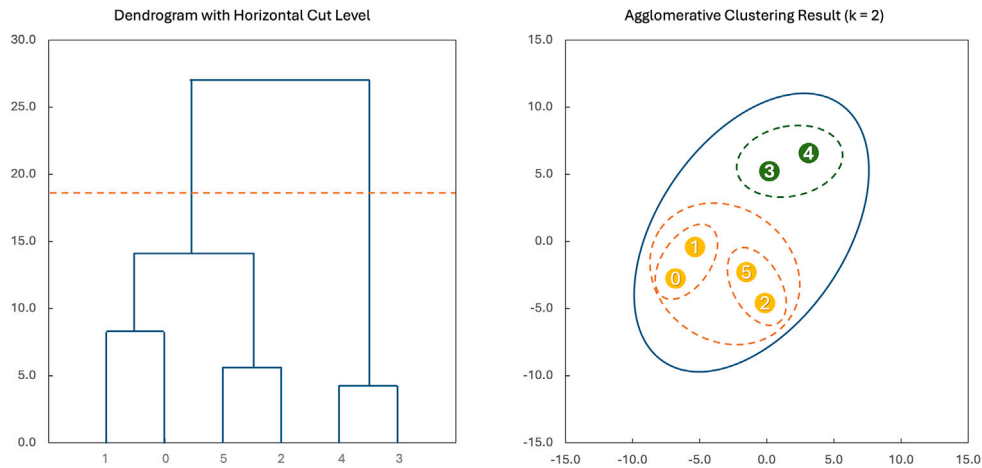


Fig. 4. Illustration of hierarchical clustering. The left panel displays a dendrogram showing the nested relationships among samples. The right panel shows clusters obtained by selecting a horizontal cut level in the dendrogram. Different cut levels correspond to different levels of granularity in cluster formation.

and require a predefined number of clusters, which can affect clustering performance.

- Hierarchical-Based Clustering (e.g., Fig. 4): The sole concept of hierarchical clustering lies in the construction and analysis of a dendrogram. A dendrogram is a tree-like structure that shows the relationships among all the data points in the system. There is no need to specify a predefined number of clusters. These algorithms can handle clusters of various shapes and sizes. Once the dendrogram is constructed, a horizontal cut through the structure defines individual clusters at the highest level in the system. Each resulting child branch below this cut represents a distinct cluster, assigning each data sample to a cluster. However, hierarchical clustering requires intensive computation, especially for large datasets, and its time complexity is often higher than that of other clustering methods. This structure is sensitive to noise and outliers, which can lead to sub-optimal clusters without the appropriate handle.
- Density-Based Clustering (e.g., Fig. 5): Density-based algorithms aim to identify clusters in high-dimensional regions of the feature space while also detecting noise points as outliers. These algorithms have high clustering efficiency, are sensitive to parameter settings, and are suitable for datasets of arbitrary shape. When spatial data density is uneven, clustering quality decreases. Additionally, more computing resources are required to process large datasets.

- Grid-Based Clustering (e.g., Fig. 6): The fundamental principle of these clustering algorithms is to partition the initial data space into a grid structure of a predetermined size for clustering. While exhibiting low time complexity, high scalability, and compatibility with parallel processing and incremental updates, these algorithms do come with considerations. The clustering outcomes are sensitive to the grid size, and the pursuit of greater computational efficiency may come at the expense of cluster quality and overall clustering accuracy.
- Model-Based Clustering (e.g., Fig. 7): The basic idea is to select a particular model for each cluster and find the best fitting for that model. Model-based clustering algorithms presume data points are generated from a probabilistic model and seek to identify the most appropriate model to explain the data distribution. Diverse and well-developed models provide means to describe data adequately, and each model has unique characteristics that may offer notable benefits in specific areas. However, overall, the models' time complexity is relatively high, the premise is not entirely true, and the clustering results depend on the chosen model parameters.

4.1.2. Deep representation-based clustering

Deep representation-based clustering (e.g., Fig. 8) integrates representation learning and clustering into a unified framework by leveraging deep neural networks to learn informative latent features from

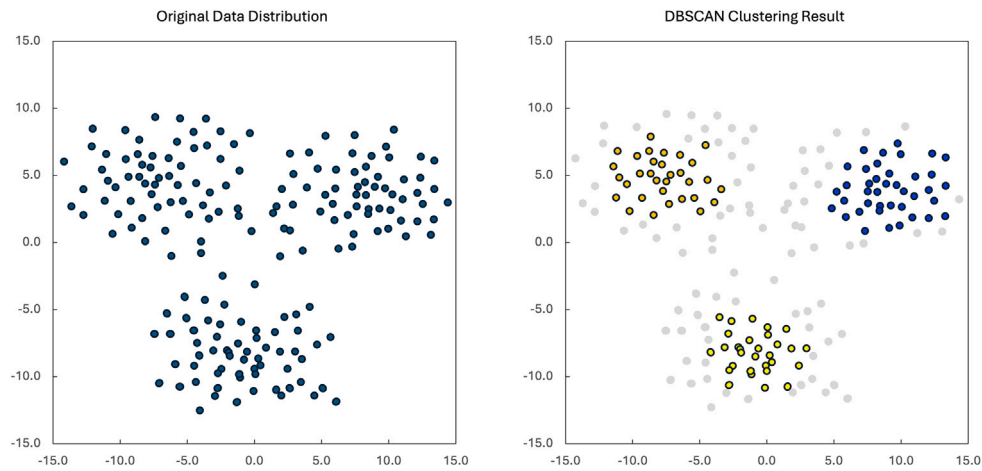


Fig. 5. Illustration of density-based clustering (DBSCAN). (a) Original data distribution. (b) Clusters identified by DBSCAN based on density connectivity. Points in low-density regions are labelled as noise.

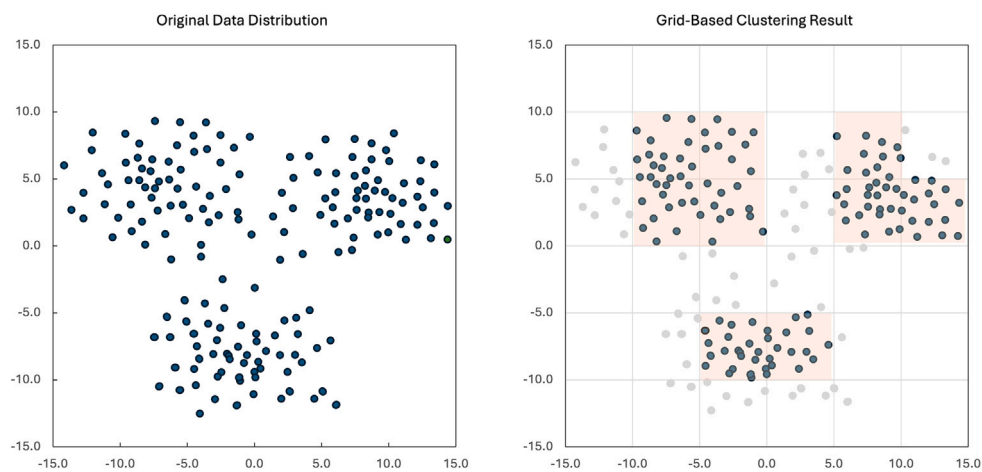


Fig. 6. Illustration of grid-based clustering. The left panel shows the original data distribution. The right panel illustrates how the feature space is partitioned into a predefined grid. Data points are aggregated within grid cells, and cells whose densities exceed a threshold are identified as dense. Clusters are formed by merging spatially adjacent dense cells according to connectivity rules, while low-density cells are treated as noise.

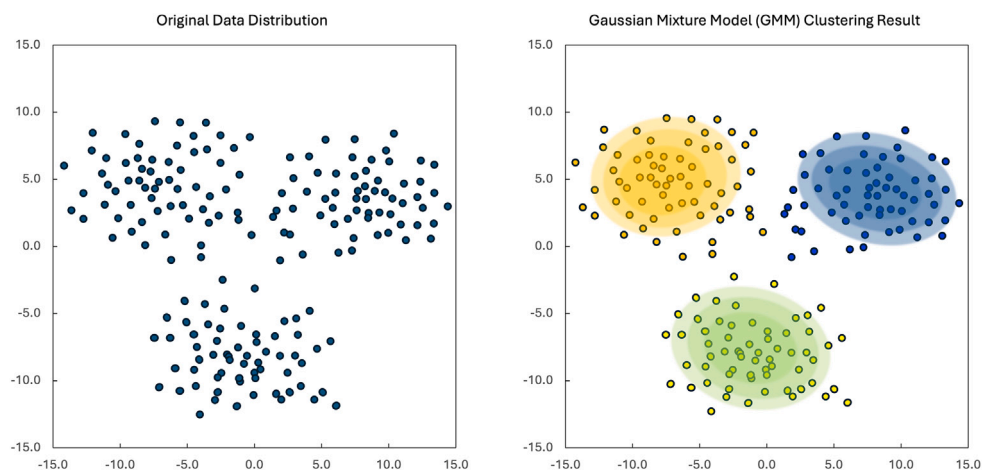


Fig. 7. Illustration of model-based clustering (Gaussian Mixture Model). The left panel shows the original data distribution. The right panel presents clusters obtained by fitting Gaussian mixture components to the data. Each cluster corresponds to a learned Gaussian distribution characterised by its mean and covariance, and cluster membership is determined probabilistically.

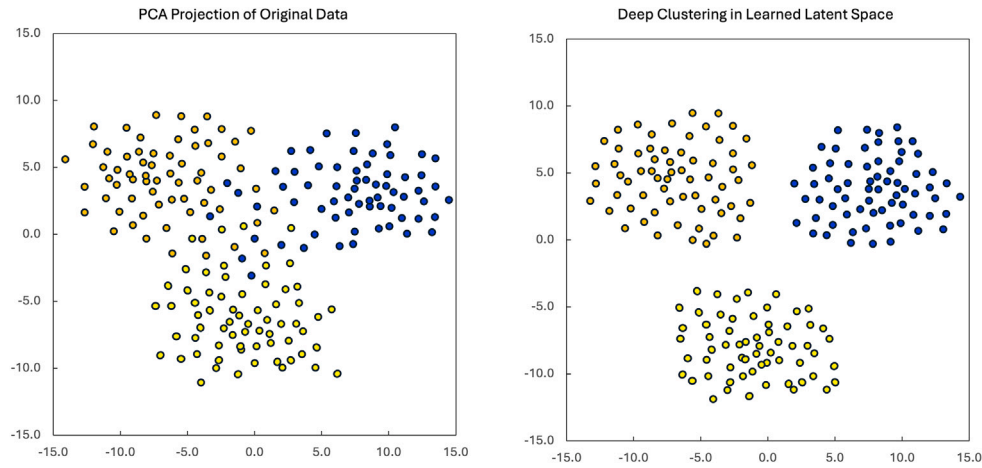


Fig. 8. Illustration of deep representation-based clustering. The left panel shows the PCA projection of the original feature space. The right panel presents clustering results in a learned latent representation (visualised via PCA). Deep models first transform the data through nonlinear feature learning, enhancing cluster separability prior to clustering.

Table 1

Algorithms classified by underlying principles and characteristics.

Category	Benefits	Limitations	Algorithm examples
Partition-based clustering	Efficient (low time complexity). Relatively simple to implement.	Sensitive to outliers. Drawn to local optima. Requires a predefined number of clusters (k).	K-Means [7,8], K-medoids [22–24], PAM [22], CLARA [25], CLARANS [26], FCM [10]
Hierarchical clustering	Does not require a predefined number of clusters. A dendrogram provides hierarchical information on clusters and subclusters.	Inefficient (high time complexity). Sensitive to noise and outliers due to dendrogram structure.	Agglomerative Clustering [27], Chameleon [28], BIRCH [29], CURE [30], Complete Linkage [31], ROCK [32], Single Linkage Clustering [31], Divisive Clustering [33]
Density-based clustering	Effective at discovering arbitrary-shaped clusters.	Decreased clustering quality for uneven spatial data density. Inefficient for large datasets.	DBSCAN [9], DENCLUE [34], HDBSCAN [35], Mean-Shift Clustering [36], OPTICS [37]
Grid-based clustering	Efficient (low time complexity). Suitable for arbitrary cluster shapes.	Clusters sensitive to grid size.	WaveCluster, CLIQUE [38], STDBSCAN [39], Grid-DBSCAN [9], STING [40]
Model-based clustering	Allows for non-spherical clusters. Produces uncertainty of cluster membership. Does not require a predefined number of clusters.	Sensitive to model assumptions and parameter initialisation.	GMM [41], Hidden Markov Models (HMM) [42], Latent Dirichlet Allocation (LDA) [43], Expectation-Maximisation Clustering (EM) [44]
Deep learning-based clustering	Learns informative latent representations. Effective for high-dimensional and nonlinear data. Improves cluster compactness and separability.	Requires large training data. High computational cost. Sensitive to network architecture and hyperparameter settings. Limited interpretability.	DEC [45], IDEC [46], VaDE [47], DCEC [48], DCN [49], Contrastive Clustering [50]

data. The basic idea of these methods is to transform high-dimensional or complex data into a low-dimensional embedding space, where cluster structures become more compact and separable. Compared with classical clustering algorithms, deep clustering approaches are more effective in handling high-dimensional, nonlinear, and unstructured data, such as text, images, and graph data, and can significantly improve clustering performance through learned feature representations. However, these methods usually require large training datasets, involve higher computational cost, depend on network architecture design and hyperparameter tuning, and may suffer from reduced interpretability compared with traditional clustering approaches.

4.2. Assignment mechanisms: Single vs. Multi

Clustering algorithms can be categorised based on how they assign data points to clusters. This results in two fundamental assignment mechanisms: single-membership assignment, where each data point belongs to only one cluster, and multiple-membership assignment, where a data point may belong to several clusters simultaneously. These mechanisms align closely with the concepts of hard and soft clustering, although they are not fully equivalent.

- Single-membership (exclusive) clustering: K-Means, Hierarchical Agglomerative Clustering, DBSCAN, OPTICS, K-Medoids, Spectral Clustering (K-way Normalised Cut), BIRCH.
- Multiple-membership (overlapping/soft) clustering: Gaussian Mixture Models (GMM), Fuzzy C-Means (FCM), Possibilistic C-Means (PCM), Possibilistic Fuzzy C-Means (PFCM), Kernelised Fuzzy C-Means (KFCM), Hierarchical Fuzzy Clustering (HFC).

It is important to note that while single membership is typical of hard clustering and multiple membership is typical of soft clustering, the two concepts are related but not identical.

4.3. Scalability and dataset capacity

Scalability is a critical factor in the practical application of clustering algorithms. One fundamental determinant of scalability is computational time complexity, which provides a theoretical framework for understanding how algorithm performance evolves with increasing dataset size. Although actual runtime depends on implementation details and hardware configurations, the commonly reported Big-O

Table 2
Typical time complexity of representative clustering paradigms.

Paradigm	Representative algorithm	Typical time complexity
Partition-based	K-Means	$O(nkid)$
Hierarchical	Agglomerative	$O(n^2) - O(n^3)$
Density-based	DBSCAN	$O(n \log n)$ (with index); $O(n^2)$ worst case
Grid-based	STING/CLIQUE	$O(n + m)$
Model-based	GMM (EM)	$O(nkid^2)$ (full covariance)
Deep clustering	DEC-style models	$O(neC_{net})$

Note: Complexities are representative and commonly reported for typical implementations. Actual runtime may vary depending on linkage strategy, indexing structures, covariance assumptions, parameter settings, data characteristics, and hardware configurations.

complexity of representative clustering paradigms offers a useful analytical reference. Table 2 summarises these typical time complexity characteristics.

Building upon this theoretical perspective, clustering algorithms differ in their practical capacity to handle datasets of varying sizes. In this section, we further classify clustering methods into three categories — small, medium, and large — according to their processing capabilities. In practice, defining what constitutes a small, medium, or large dataset can be somewhat subjective and task-dependent. Based on an overview of the open-source library for Python¹ and existing work, we categorise algorithms into three groups according to their processing capabilities, as illustrated in Table 3.

Small Dataset: Typically, a small dataset might contain a few hundred to a few thousand instances. It is a scale where the entire dataset can be easily loaded into memory and processed without significant computational resources.

Medium Dataset: A medium-sized dataset could range from a few thousand to tens of thousands of instances. It might require more sophisticated algorithms and computational resources compared to a small dataset, but it is still manageable.

Large Dataset: Large datasets typically contain hundreds of thousands to millions (or more) instances. Handling such datasets often requires specialised algorithms, distributed computing, or parallel processing due to the sheer volume of data.

4.4. Predefined cluster numbers

Not all clustering algorithms require the number of clusters to be specified in advance. From this perspective, clustering methods can be broadly divided into two categories:

Algorithms requiring a predefined number of clusters: These methods take the desired number of clusters, K , as an input parameter and optimise an objective function under this constraint. Typical examples include prototype-based and model-based algorithms such as K-Means, K-Medoids, Spectral Clustering (K-way Normalised Cut), Gaussian Mixture Models (GMM), Fuzzy C-Means (FCM), Possibilistic C-Means (PCM), Possibilistic Fuzzy C-Means (PFCM), and Kernelised Fuzzy C-Means (KFCM).

Algorithms that do not require a predefined number of clusters: These methods infer the number of clusters from the data or provide a hierarchical structure from which an appropriate cut can be chosen. Representative examples include density-based methods such as DBSCAN and OPTICS, as well as hierarchical agglomerative clustering and BIRCH. In hierarchical clustering, for instance, the algorithm produces a dendrogram that captures cluster structure at multiple levels of granularity; the analyst can then select a suitable number of clusters by choosing an appropriate cut level.

¹ <https://scikit-learn.org/stable/modules/clustering.html#overview-of-clustering-methods>.

It is important to note that the distinction between methods that require a predefined number of clusters and those that do not is conceptually distinct from that between hard and soft clustering. Both hard (exclusive) and soft (fuzzy or probabilistic) clustering algorithms may require the number of clusters to be specified in advance.

A major practical challenge for algorithms that require a predefined K is determining the optimal number of clusters. Clustering results can vary substantially with different choices of K , affecting the granularity of the discovered patterns and the interpretability and usefulness of the resulting groups in real-world applications. Consequently, selecting an appropriate number of clusters is a crucial step in the clustering process. Several commonly used methods for estimating the optimal number of clusters include:

- **Elbow method**²: The clustering algorithm is run for a range of candidate values of K , and the within-cluster sum of squares (inertia) is plotted against K . The “elbow” point, where the marginal gain (reduction in inertia) begins to diminish, is often taken as a reasonable choice of K . The elbow method has been widely used in practice [53–55].
- **Silhouette score**³: The silhouette coefficient measures how similar an object is to its own cluster compared with other clusters. For each candidate value of K , an overall silhouette score is computed, and the value of K that maximises this score is typically preferred. Numerous studies employ the silhouette score to evaluate the quality of clustering solutions [56–58].
- **Gap statistic** [59]: The gap statistic compares the within-cluster dispersion for the observed data to that expected under a suitable reference (null) distribution. The optimal number of clusters is chosen as the smallest K that yields a near-maximum gap between the observed and reference dispersions. Several publications have applied the gap statistic to estimate the number of clusters [60–62].
- **Dendrogram analysis in hierarchical clustering** [63]: When using hierarchical clustering, the dendrogram provides a visual summary of how clusters are merged or split at different levels. By examining the dendrogram and identifying large vertical gaps or meaningful partitions, an appropriate cut level — and hence the number of clusters — can be selected. This approach has been employed in various applications [64,65].

4.5. Area of application

Clustering algorithms are widely applied across diverse domains, presenting unique data characteristics and analytical objectives. Table 4 summarises key application areas and commonly used clustering techniques in each field.

In data mining, algorithms such as K-Means and DBSCAN are widely adopted for their computational efficiency and scalability. These methods are particularly effective when processing large-scale, high-dimensional datasets, such as those found in customer segmentation or anomaly detection.

In image analysis, techniques like Spectral Clustering and Mean Shift are favoured due to their ability to handle complex image structures and capture spatial continuity. Spectral Clustering is well-suited for tasks involving irregular shapes or non-convex clusters, while Mean Shift adapts well to varying image densities without needing a predefined number of clusters. With the rapid advancement of deep learning, deep representation-based clustering has become increasingly prominent in image and medical imaging applications. By learning discriminative latent feature spaces through neural networks prior to clustering, these approaches enhance separability and improve performance on high-dimensional visual data.

² [https://en.wikipedia.org/wiki/Elbow_method_\(clustering\)](https://en.wikipedia.org/wiki/Elbow_method_(clustering)).

³ [https://en.wikipedia.org/wiki/Silhouette_\(clustering\)](https://en.wikipedia.org/wiki/Silhouette_(clustering)).

Table 3
Algorithms classified by dataset capacity.

Dataset	Algorithm examples	Comments
Small (Up to a few thousand instances)	K-Means, DBSCAN, Hierarchical clustering	These algorithms are often efficient and can run on standard machines, thereby avoiding the need for extensive computational resources.
Medium (Several thousand to hundreds of thousands of instances)	Optimised K-Means, GMM, Optimised Agglomerative clustering, Mean-Shift clustering	The algorithms may utilise more sophisticated techniques to deal with the increased size of the dataset.
Large [51,52] (Hundreds of thousands of instances or more)	Mini-Batch K-Means, BIRCH, DBSCAN (optimised for large datasets), Hierarchical clustering (optimised for large datasets)	These algorithms are tailored for parallel processing to efficiently tackle computational challenges in handling large datasets.

Table 4
Areas of application of clustering algorithms.

Area of application	Algorithm examples
Data mining	K-Means, DBSCAN, HDBSCAN
Image analysis	Spectral Clustering, Mean Shift
Pattern recognition	Hierarchical Clustering, K-Means, SpectralNet
Information retrieval	Hierarchical Clustering, K-Means, DBSCAN
Bioinformatics	Hierarchical Clustering, Spectral Clustering
Image reconstruction	K-Means, Superpixel, Expectation Maximisation
Network analysis	Hierarchical Clustering, K-Means, DBSCAN, AutoClass

In pattern recognition, both Hierarchical Clustering and K-Means are commonly used. Hierarchical Clustering is valuable when a nested structure of patterns is present, allowing analysis at multiple levels of granularity. K-Means remains useful for recognising fixed patterns where the number of categories is known or can be estimated. Deep clustering methods further extend these capabilities by integrating representation learning with clustering objectives, enabling improved performance in complex recognition tasks.

In information retrieval, algorithms like Hierarchical Clustering, K-Means, and DBSCAN are used to group documents, keywords, or users. These clustering methods enhance search performance by organising information into semantically meaningful clusters, which can then support indexing, filtering, and recommendation systems.

In bioinformatics, Hierarchical Clustering and Spectral Clustering are widely used to analyse gene expression data, protein interactions, and sequence similarities. These methods help reveal functional groupings and hidden biological structures without assuming linear boundaries. Recently, deep clustering approaches have also been explored in genomics and biomedical data analysis, particularly for uncovering nonlinear structures in large-scale omics datasets.

In image formation, clustering plays a key role in simplifying and segmenting visual data. K-Means is often used for colour quantisation, Superpixel techniques preserve local structures for more coherent regions, and Expectation Maximisation (EM) helps in modelling pixel distributions probabilistically.

In network analysis, where the relationships between nodes (such as users or devices) are of interest, various clustering methods are employed. Hierarchical Clustering, K-Means, DBSCAN, and Bayesian methods such as AutoClass are utilised to detect communities, roles, or functional modules, depending on the network’s structure and scale.

It is important to note that these pairings are not absolute; algorithm selection often depends on the actual scenario, such as the dataset’s size, noise level, shape complexity, and interpretability needs. As new clustering methods emerge, such as AutoClass [66] and Superpixel-based techniques [67], they expand the range of tasks that can be effectively addressed.

5. Evaluation metrics

Although most clustering datasets lack ground truth labels, various evaluation methods have been developed to assess clustering quality. These metrics are critical in guiding the development, selection, and optimisation of clustering algorithms, making the evaluation process

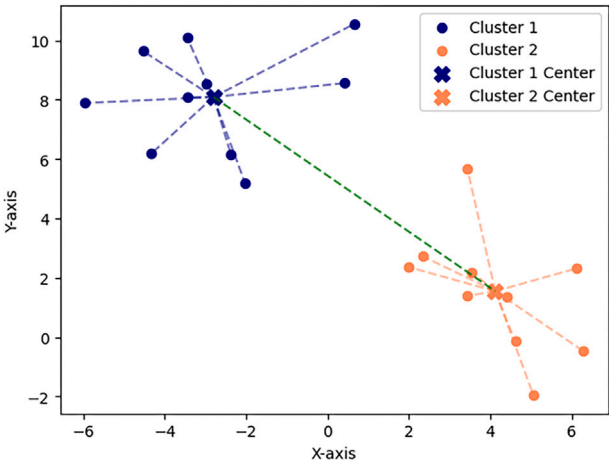


Fig. 9. Example of K-Means clustering with two clusters, illustrating two different types of distance between data points.

more systematic, objective, and informative. In the clustering process, data samples transform into vectors in a high-dimensional space. The distances between these vectors intricately reflect the overall similarity, incorporating all relevant features within the data samples. Therefore, “distances between points” are pivotal in forming clusters and serve as a common standard for evaluating clustering performance. This concept refers to numerical measures of dissimilarity or similarity between individual data points, often calculated based on data features. Typical metrics include Euclidean distance, Manhattan distance, and other similarity measures, such as cosine similarity, for text data. Fig. 9 illustrates an example of K-Means clustering, showcasing the distances between data points. It represents an idealised scenario with well-separated clusters; however, real-world applications often involve more complex and overlapping data distributions. Two types of distances are depicted: one represents the distance between data samples within the cluster (blue and orange dashed lines), while the other indicates the distance between different cluster centers (green dashed line). The distance between data points within a cluster is considerably smaller than between data points and the centres of other clusters.

By emphasising the importance of evaluation metrics and the role of point-to-point distances, the process of assessing and improving clustering algorithms becomes more robust and meaningful. Evaluation metrics are categorised into internal and external types. Internal metrics assess the quality of clusters based solely on inherent data information, focusing on factors like the compactness (cohesion) of data samples within a cluster and the distinctiveness (separation) between clusters. Examples include the Silhouette Coefficient, Davies–Bouldin Index, Dunn’s Index, and Inertia (for K-Means). External metrics require true class labels, evaluating alignment with known labels using precision, recall, and mutual information. Such as the Adjusted Rand Index (ARI), Normalised Mutual Information (NMI), Fowlkes–Mallows Index, Precision, Recall, and F1 Score. Internal metrics focus on intrinsic properties, while external metrics rely on ground truth to evaluate algorithm accuracy. The choice depends on data availability and clustering goals.

5.1. Internal evaluation metrics

Internal evaluation metrics for clustering algorithms assess the quality of clusters solely from the data's intrinsic characteristics and the clustering algorithm's results, without using external information, such as ground truth labels. Various aspects of clustering quality can be quantified using these metrics, such as compactness, separation, and variance. When choosing and interpreting these metrics, it is essential to consider the specific characteristics of the data and the goals of the clustering task.

5.1.1. Silhouette Score

The Silhouette Score is a widely used internal evaluation metric for assessing the quality of clusters produced by unsupervised clustering algorithms. It measures how well-separated clusters are and provides an indication of the appropriateness of the clustering solution. The Silhouette Score for a single data point is calculated as the difference between the average distance from the data point to other points in the same cluster (a) and the average distance from the data point to points in the nearest neighbouring cluster (b), divided by the maximum of a and b. For a data point i , the formula is:

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (1)$$

The Silhouette Score ranges from -1 to 1 , with higher values indicating better-defined clusters. The Score close to 1 indicates that the data point is well matched to its cluster and poorly matched to neighbouring clusters, suggesting a good clustering. A Silhouette Score around 0 indicates overlapping clusters, and a Silhouette Score close to -1 indicates that the data point may be assigned to the wrong cluster. The overall Silhouette Score for a clustering solution is the average of the Silhouette Scores for all data points.

5.1.2. Davies–Bouldin Index

The Davies–Bouldin Index (DBI) is a metric used for evaluating the quality of clustering solutions. It measures compactness and cluster separation, aiming to identify well-separated clusters. The formula of Davies–Bouldin Index is:

$$DBI = \frac{1}{n_c} \sum_{i=1}^{n_c} \max_{j \neq i} \left(\frac{avg_radius_i + avg_radius_j}{distance(c_i, c_j)} \right) \quad (2)$$

where: n_c is the number of clusters, c_i and c_j are the centroids of clusters i and j , avg_radius_i and avg_radius_j are the average distances from the centroid to the points in cluster i and j , $distance(c_i, c_j)$ is the distance between the centroids of clusters i and j .

The Davies–Bouldin Index is based on the Euclidean distance, which assumes clusters are spherical and may not perform well with irregularly shaped clusters. Also, the formula assumes equal-sized clusters, so it might not be suitable for datasets with imbalanced cluster sizes. It may not perform well if the true number of clusters is unknown or if the clustering algorithm produces a different number of clusters. Despite these limitations, the Davies–Bouldin Index can be a useful tool when its assumptions align with the data's characteristics and the goals of the clustering analysis.

5.1.3. Dunn's Index

Dunn's Index [68] is an internal cluster validity measure designed to identify compact and well-separated clusters. The index favours clustering solutions in which the minimum separation between different clusters is large, and the maximum diameter within any cluster is small.

To avoid notation ambiguity, we define:

- $\delta(C_i, C_j)$ as the *inter-cluster distance* between clusters C_i and C_j , typically computed as the minimum pairwise distance between points in the two clusters:

$$\delta(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y),$$

- $\Delta(C_k)$ as the *intra-cluster diameter* of cluster C_k :

$$\Delta(C_k) = \max_{x, y \in C_k} d(x, y).$$

Using these definitions, Dunn's Index D is expressed as:

$$D = \frac{\min_{1 \leq i < j \leq K} \delta(C_i, C_j)}{\max_{1 \leq k \leq K} \Delta(C_k)}.$$

A higher value of D indicates better clustering quality, corresponding to better-separated clusters (large numerator) and tighter, more compact clusters (small denominator). Because the index emphasises extreme values (minimum separation and maximum diameter), it tends to be sensitive to noise and outliers.

5.2. External evaluation metrics (with ground truth)

5.2.1. Adjusted Rand Index (ARI)

The Adjusted Rand Index (ARI) is a widely used metric in cluster analysis and machine learning for evaluating the similarity between two clustering solutions. It measures the agreement between the true class labels and the labels assigned by a clustering algorithm while correcting for chance. The formula of ARI is:

$$ARI = \frac{RI - \text{Expected } RI}{\text{Max}(RI) - \text{Expected } RI} \quad (3)$$

where RI is the Rand Index, which measures the proportion of agreements (both in the same cluster or both in different clusters) between the true and predicted clusterings. $\text{Expected } RI$ is the expected Rand Index under the assumption of random clustering. It represents the expected value of RI when clustering is performed randomly. The $\text{max}(RI)$ term in the denominator represents the maximum possible Rand Index, which normalises the ARI to the range $[-1, 1]$.

Although the Adjusted Rand Index (ARI) is widely used, it has several well-known limitations. First, ARI is sensitive to the distribution of cluster sizes: it tends to favour solutions that produce a similar number of clusters or clusters of comparable sizes, which may bias the evaluation when the true clusters are highly imbalanced. Second, ARI can be unstable in the presence of noise or many small clusters, as minor perturbations in assignment may lead to disproportionately large changes in the score. Additionally, ARI requires a one-to-one correspondence between predicted and true clusters, making it less suitable for settings involving overlapping clusters or soft membership assignments. Finally, because ARI evaluates cluster labels relative to ground truth, it cannot assess clustering quality in fully unsupervised scenarios, where no reference partition is available.

5.2.2. Normalised Mutual Information (NMI)

Normalised Mutual Information provides a quantitative measure of the agreement between the true class labels and the labels assigned by a clustering algorithm, taking into account both precision and recall, offering a balanced perspective on the quality of the clustering solution.

The formula of NMI is:

$$NMI = \frac{2 \cdot I(Y; C)}{H(Y) + H(C)} \quad (4)$$

where Y is the set of true class labels, C is the set of cluster labels assigned by the algorithm, $I(Y; C)$ is the mutual information between Y and C , and $H(Y)$ and $H(C)$ are the entropies of Y and C , respectively. NMI ranges from 0 to 1 , where 0 indicates no mutual information, and 1 implies perfect agreement between the true and predicted labels. A higher NMI suggests a better clustering solution for capturing the underlying class structure. NMI is a commonly used metric in most clustering solutions, as it accounts for both homogeneity and completeness in clustering evaluation. Normalisation helps in comparing NMI across datasets of different sizes.

NMI has several known limitations. It tends to overestimate similarity when a clustering produces many small clusters, as mutual

information can increase artificially with over-partitioning. NMI is also sensitive to highly imbalanced cluster sizes, which may distort the comparison between two partitions. In addition, it assumes crisp, non-overlapping clusters and is not directly suitable for fuzzy or overlapping assignments. Finally, NMI requires a ground-truth partition and therefore cannot be used in fully unsupervised evaluation.

6. Discussion

In recent years, the focus of clustering research has shifted from purely advancing algorithmic formulations to developing application-driven solutions tailored to specific domains. This evolution is largely driven by the diverse, increasingly complex data challenges arising in modern fields. Researchers now actively investigate how clustering algorithms can be adapted and optimised to meet the unique requirements of areas such as bioinformatics [69–71], healthcare [57,72,73], natural language processing [74,75], image and video analysis [76,77], social network analysis [78–80], cyber security [81–84], and anomaly detection [38,85–87]. This shift has led to substantial progress in domain-specific clustering methodologies, reflecting a broader trend towards customisation rather than generic algorithmic improvements.

A noticeable surge in clustering-related publications occurred during the COVID-19 pandemic, particularly in medical imaging and healthcare applications from 2021 to 2022. Beyond application-driven growth, the rapid advancement of deep learning has fundamentally reshaped clustering research. In response to this paradigm shift, we have explicitly incorporated deep representation-based clustering as a dedicated subcategory within our taxonomy.

Increasingly, neural networks and representation-learning techniques are embedded into clustering frameworks to learn task-adaptive latent spaces prior to cluster assignment [88–90]. This integration improves separability in high-dimensional, complex data scenarios. As a result, deep clustering and hybrid approaches — combining multiple clustering paradigms or integrating clustering with other machine learning strategies — have gained substantial momentum. These developments reflect a broader methodological transition from distance-based partitioning towards representation-driven clustering paradigms.

Despite decades of development, classical algorithms such as K-Means remain dominant. Its simplicity, efficiency, and strong baseline performance continue to make it a standard reference method for comparison in new research [91–94]. Efforts to refine, accelerate, or extend K-Means illustrate its enduring influence in the field.

Across the literature, a central challenge persists: determining the optimal number of clusters. Existing algorithms often produce ambiguous or poorly defined groupings, in which clusters may not correspond to meaningful or interpretable structures. In some cases, multiple plausible partitions exist, resulting in discrepancies between algorithmic output and human judgment. These issues highlight the inherent difficulty of selecting a universally superior clustering method. Performance varies widely depending on data characteristics, domain constraints, and interpretability requirements. As a result, clustering remains highly task-dependent, with no single method outperforming others across all scenarios.

To address this, our review classifies clustering algorithms from multiple perspectives — such as assignment mechanisms, scalability, supervision level, and predefined cluster requirements — providing practical guidance for selecting suitable clustering techniques for diverse real-world applications.

CRediT authorship contribution statement

Hui Yin: Writing – review & editing, Writing – original draft, Visualization, Investigation. **Amir Aryani:** Writing – review & editing, Supervision, Conceptualization. **Stephen Petrie:** Writing – review & editing. **Aishwarya Nambissan:** Writing – review & editing, Writing – original draft. **Aland Astudillo:** Writing – review & editing. **Shengyuan Cao:** Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve language. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

References

- [1] Wagstaff Kiri, Cardie Claire, Rogers Seth, Schrödl Stefan, et al. Constrained k-means clustering with background knowledge. In: *Icml*. vol. 1, 2001, p. 577–84.
- [2] Benkhalfia Mohammed, Bensaid Amine, Mouradi Abdelhak. Text categorization using the semi-supervised fuzzy c-means algorithm. In: *18th international conference of the North American fuzzy information processing society-NAFIPS* (cat. no. 99TH8397). IEEE; 1999, p. 561–5.
- [3] Chen Weifu, Feng Guocan. Spectral clustering: A semi-supervised approach. *Neurocomput* 2012;77(1):229–42. <http://dx.doi.org/10.1016/j.neucom.2011.09.002>.
- [4] A deep-based gaussian mixture model algorithm for large-scale many objective optimization. *Appl Soft Comput* 2025;172:112874. <http://dx.doi.org/10.1016/j.asoc.2025.112874>, URL <https://www.sciencedirect.com/science/article/pii/S1568494625001851>.
- [5] Yin Hui, Song Xiangyu, Yang Shuiqiao, Li Jianxin. Sentiment analysis and topic modeling for covid-19 vaccine discussions. *World Wide Web* 2022;25(3):1067–83. <http://dx.doi.org/10.1007/s11280-022-01029-y>.
- [6] Yin Hui, Yang Shuiqiao, Li Jianxin. Detecting topic and sentiment dynamics due to covid-19 pandemic using social media. In: Yang Xiaochun, Chang-Dong Wang MdSaiful Islam, Zhang Zheng, editors. *Advanced data mining and applications*. Cham: Springer International Publishing; 2020, p. 610–23.
- [7] Lloyd Stuart P. Least squares quantization in pcm. *IEEE Trans Inform Theory* 1982;28(2):129–37. <http://dx.doi.org/10.1109/TIT.1982.1056489>.
- [8] Forgy Edward W. Cluster analysis of multivariate data: Efficiency versus interpretability of classifications. *Biometrics* 1965;21(3):768–9.
- [9] Ester Martin, Kriegel Hans-Peter, Sander Jörg, Xu Xiaowei. A density-based algorithm for discovering clusters in large spatial databases with noise. In: *Proceedings of the second international conference on knowledge discovery and data mining*. KDD-96, Portland, Oregon, USA: AAAI Press; 1996, p. 226–31, doi:10.1.1.121.9220.
- [10] Bezdek James C, Ehrlich Robert, Fcm William Full. The fuzzy c-means clustering algorithm. *Comput Geosci* 1984;10(2–3):191–203.
- [11] Xu Rui, Wunsch D. Survey of clustering algorithms. *IEEE Trans Neural Netw* 2005;16(3):645–78. <http://dx.doi.org/10.1109/TNN.2005.845141>.
- [12] Xu Dongkuan, Tian Yingjie. A comprehensive survey of clustering algorithms. *Ann Data Sci* 2015;2:165–93. <http://dx.doi.org/10.1007/s40745-015-0040-1>.
- [13] Ezugwu Absalom E, Ikotun Abiodun M, Oyelade Olaide O, Abualigah Laith, Agushaka Jeffery O, Eke Christopher I, Akinyelu Andronicus A. A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects. *Eng Appl Artif Intell* 2022;110:104743. <http://dx.doi.org/10.1016/j.engappai.2022.104743>.

- [14] Bora Dibya Jyoti, Gupta Anil Kumar. A comparative study between fuzzy clustering algorithm and hard clustering algorithm. 2014, CoRR abs/1404.6059 URL <http://arxiv.org/abs/1404.6059>.
- [15] Sisodia Deepti, Singh Lokesh, Sisodia Sheetal, Saxena Khushboo. Clustering techniques: a brief survey of different clustering algorithms. *Int J Latest Trends Eng Technol (IJLTET)* 2012;1(3):82–7.
- [16] Rodriguez Mayra Z, Comin Cesar H, Casanova Dalcimar, Bruno Odemir M, Amancio Diego R, da F Costa Luciano, Rodrigues Francisco A. Clustering algorithms: A comparative approach. *PLoS One* 2019;14(1):e0210236. <http://dx.doi.org/10.1371/journal.pone.0210236>.
- [17] Zhou Sheng, Xu Hongjia, Zheng Zhuonan, Chen Jiawei, Bu Jiajun, Wu Jia, Wang Xin, Zhu Wenwu, Ester Martin, et al. A comprehensive survey on deep clustering: Taxonomy, challenges, and future directions. 2022, <http://dx.doi.org/10.48550/arXiv.2206.07579>, arXiv preprint arXiv:2206.07579.
- [18] Sajana T, Rani C Sheela, Narayana KV. A survey on clustering techniques for big data mining. *Indian J Sci Technol* 2016;9(3):1–12. <http://dx.doi.org/10.17485/ijst/2016/v9i3/75971>.
- [19] Berkhin Pavel. A survey of clustering data mining techniques. In: Kogan Jacob, Nicholas Charles, Tebouille Marc, editors. *Grouping multidimensional data: recent advances in clustering*. Berlin, Heidelberg: Springer; 2006, p. 25–71. http://dx.doi.org/10.1007/3-540-28349-8_2.
- [20] Kumar Dheeraj, James C Bezdek, Palaniswami Marimuthu, Rajasegarar Sutharshan, Leckie Christopher, hybrid approach to clustering in big data Timothy Craig Havens A. *IEEE Trans Cybern* 2016;46(10):2372–85. <http://dx.doi.org/10.1109/TCYB.2015.2477416>.
- [21] You Yi Zi, Pan YunRong, Ma Zhi, Zhang Li, Xiao Shuo, Zhang Dan Dan, Dang Shijun, Zhao Shuang Ru, Wang Pei, Dong Ai-Jun, et al. Applying hybrid clustering in pulsar candidate sifting with multi-modality for fast survey. *Res Astron Astrophys* 2023. <http://dx.doi.org/10.1088/1674-4527/ad0c28>.
- [22] Kaufman Leonard, Peter J Rousseeuw. *Partitioning around medoids (program PAM)*. Hoboken, NJ, USA: John Wiley & Sons, Inc.; 1990, <http://dx.doi.org/10.1002/9780470316801.ch2>.
- [23] Schubert Erich, Peter J Rousseeuw. Fast and eager k-medoids clustering: O(k) runtime improvement of the pam, clara, and clarans algorithms. *Inf Syst* 2021;101:101804. <http://dx.doi.org/10.1016/j.is.2021.101804>.
- [24] Rousseeuw LKJP, Kaufman P. Clustering by means of medoids. In: *Proceedings of the statistical data analysis based on the 11 norm conference, neuchatel, Switzerland*. vol. 31, 1987.
- [25] Kaufman Leonard, Peter J Rousseeuw. *Finding groups in data: an introduction to cluster analysis*. Hoboken, NJ: John Wiley & Sons; 2009.
- [26] Ng Raymond T, Clarans Jiawei Han. A method for clustering objects for spatial data mining. *IEEE Trans Knowl Data Eng* 2002;14(5):1003–16. <http://dx.doi.org/10.1109/TKDE.2002.1033770>.
- [27] Johnson Stephen C. Hierarchical clustering schemes. *Psychometrika* 1967;32(3):241–54.
- [28] Karypis George, Han Eui-Hong, Kumar Vipin. Chameleon: Hierarchical clustering using dynamic modeling. *Computer* 1999;32(8):68–75.
- [29] Zhang T, Ramakrishnan R, Livny M. BIRCH: An efficient data clustering method for very large databases. In: *Proceedings of the 1996 ACM SIGMOD international conference on management of data*. SIGMOD; 1996, p. 103–14. <http://dx.doi.org/10.1145/233269.233324>.
- [30] Guha Sudipto, Rastogi Rajeev, Shim Kyuseok. CURE: An efficient clustering algorithm for large databases. *Inf Syst* 1998;26(1):35–58. [http://dx.doi.org/10.1016/S0306-4379\(01\)00008-4](http://dx.doi.org/10.1016/S0306-4379(01)00008-4).
- [31] Sneath PH, Sokal RR. In: Freeman WH, editor. *Numerical taxonomy: the principles and practice of numerical classification*. San Francisco, CA; 1973.
- [32] Guha Sudipto, Rastogi Rajeev, Shim Kyuseok. Rock: A robust clustering algorithm for categorical attributes. *Inf Syst* 2000;25(5):345–66.
- [33] Savaresi Sergio M, Boley Daniel L, Bittanti Sergio, Gazzaniga Giovanna. Cluster selection in divisive clustering algorithms. In: *Proceedings of the 2002 SIAM international conference on data mining*. SIAM; 2002, p. 299–314.
- [34] Hinneburg Alexander, Keim Daniel A. An efficient approach to clustering in large multimedia databases with noise. In: *Knowledge discovery and datamining. KDD'98*, 1998, p. 58–65.
- [35] Campello Ricardo JGB, Moulavi Davoud, Sander Jörg. *Density-based clustering based on hierarchical density estimates*. In: *Pacific-Asia conference on knowledge discovery and data mining*. Springer; 2013, p. 160–72.
- [36] Comaniciu Dorin, Meer Peter. Mean shift: A robust approach toward feature space analysis. *IEEE Trans Pattern Anal Mach Intell* 2002;24(5):603–19.
- [37] Ankerst Michael, Breunig Markus, Kriegel Hans-Peter, Sander Jörg. Optics: Ordering points to identify the clustering structure.. In: *Proceedings of the 1999 ACM SIGMOD international conference on management of data*. vol. 28, ACM; 1999, p. 49–60.
- [38] Agrawal Rakesh, Gehrke Johannes, Gunopulos Dimitrios, Raghavan Prabhakar. Automatic subspace clustering of high dimensional data for data mining applications. In: *Proceedings of the 1998 ACM SIGMOD international conference on management of data*. vol. 27, ACM; 1998, p. 94–105.
- [39] Birant Derya, Kut Alp. St-dbscan: An algorithm for clustering spatial-temporal data. *Data Knowl Eng* 2007;60(1):208–21. <http://dx.doi.org/10.1016/j.datak.2006.01.013>.
- [40] Wang Wang-Chien, Yang Jiong, Muntz Richard. Sting: a statistical information grid approach to spatial data mining. In: *Vldb*. 1997, p. 186–95.
- [41] Rasmussen Carl Edward. The infinite gaussian mixture model. In: *Advances in neural information processing systems*. vol. 12, 1999, p. 554–60.
- [42] Rabiner Lawrence, Juang Biinghwang. An introduction to hidden markov models. *IEEE Assp Mag* 1986;3(1):4–16.
- [43] Blei David M, Ng Andrew Y, Jordan Michael I. Latent dirichlet allocation. *J Mach Learn Res* 2003;3(Jan):993–1022.
- [44] Dempster Arthur P, Laird Nan M, Rubin Donald B. Maximum likelihood from incomplete data via the em algorithm. *J R Stat Soc Ser B Stat Methodol* 1977;39(1):1–22.
- [45] Xie Junyuan, Girshick Ross, Farhadi Ali. Unsupervised deep embedding for clustering analysis. In: *Proceedings of the 33rd international conference on machine learning*. ICML, 2016, p. 478–87. <http://dx.doi.org/10.48550/arXiv.1511.06335>.
- [46] Guo Xifeng, Gao Long, Liu Xinwang, Yin Jianping. Improved deep embedded clustering with local structure preservation. In: *Proceedings of the 26th international joint conference on artificial intelligence*. IJCAI, 2017, p. 1753–9. <http://dx.doi.org/10.24963/ijcai.2017/243>.
- [47] Jiang Zhuxi, Zheng Yin, Tan Huachun, Tang Bangsheng, Zhou Hanning. Variational deep embedding: An unsupervised and generative approach to clustering. In: *Proceedings of the 26th international joint conference on artificial intelligence*. IJCAI, 2017, p. 1965–72. <http://dx.doi.org/10.24963/ijcai.2017/273>.
- [48] Guo Xifeng, Liu Xinwang, Zhu En, Yin Jianping. Deep clustering with convolutional autoencoders. In: *International conference on neural information processing*. 2017b, p. 373–82. http://dx.doi.org/10.1007/978-3-319-70096-0_39.
- [49] Yang Bo, Fu Xiao, Sidiropoulos Nicholas D, Hong Mingyi. Towards k-means-friendly spaces: Simultaneous deep learning and clustering. In: *Proceedings of the 34th international conference on machine learning*. ICML, 2017, <http://dx.doi.org/10.48550/arXiv.1610.04794>.
- [50] Li Yunfan, Hu Peng, Liu Zitao, Peng Dezhong, Zhou Joey Tianyi, clustering Xi Peng Contrastive. *Proceedings of the AAAI conference on artificial intelligence*. vol. 35, 2021, p. 8547–55. <http://dx.doi.org/10.1609/aaai.v35i10.17037>, (10) URL <https://ojs.aaai.org/index.php/AAAI/article/view/17037>.
- [51] Shirkhorshidi Ali Seyed, Aghabozorgi Saeed, Wah Teh Ying, Herawan Tutut. Big data clustering: a review.. In: *Computational science and its applications-ICCSA 2014: 14th international conference, Guimarães, Portugal, June 30–July 3, 2014, proceedings, part v 14*. Springer; 2014, p. 707–20. http://dx.doi.org/10.1007/978-3-319-09156-3_49.
- [52] Kurasova Olga, Marcinkevicius Virginijus, Medvedev Viktor, Rapecka Aurimas, Stefanovic Pavel. Strategies for big data clustering. In: *2014 IEEE 26th international conference on tools with artificial intelligence*. IEEE; 2014, p. 740–7. <http://dx.doi.org/10.1109/ICTAI.2014.115>.
- [53] Syakur MA, Khotimah BK, Rochman EMS, Satoto Budi Dwi. Integration k-means clustering method and elbow method for identification of the best customer profile cluster.. In: *IOP conference series: materials science and engineering*. vol. 336, IOP Publishing; 2018, 012017.
- [54] Bholowalia Purnima, Kumar Arvind. Elb-means: A clustering technique based on elbow method and k-means in wsn. *Int J Comput Appl* 2014;105(9).
- [55] Marutho Dhendra, Handaka Sunarna Hendra, Wijaya Ekaprana, et al. The determination of cluster number at k-mean using elbow method and purity evaluation on headline news. In: *2018 international seminar on application for technology of information and communication*. IEEE; 2018, p. 533–8. <http://dx.doi.org/10.1109/ISEMANTIC.2018.8549751>.
- [56] Shutaywi Meshal, Kachouie Nezamoddin N. Silhouette analysis for performance evaluation in machine learning with applications to clustering. *Entropy* 2021;23(6):759. <http://dx.doi.org/10.3390/e23060759>.
- [57] Ogbuabor Godwin, Ugwoke FN. Clustering algorithm for a healthcare dataset using silhouette score value. *Int J Comput Sci Inf Technol* 2018;10(2):27–37. <http://dx.doi.org/10.5121/ijcsit.2018.10203>.
- [58] Shahapure Ketan Rajshekhar, Nicholas Charles. Cluster quality analysis using silhouette score. In: *2020 IEEE 7th international conference on data science and advanced analytics*. DSAA, IEEE; 2020, p. 747–8. <http://dx.doi.org/10.1109/DSAA49011.2020.00096>.
- [59] Tibshirani Robert, Walther Guenther, Hastie Trevor. Estimating the number of clusters in a data set via the gap statistic. *J R Stat Soc Ser B Stat Methodol* 2001;63(2):411–23.
- [60] Yan Mingjin, Ye Keying. Determining the number of clusters using the weighted gap statistic. *Biometrics* 2007;63(4):1031–7. <http://dx.doi.org/10.1111/j.1541-0420.2007.00784.x>.
- [61] El-Mandouh Amira M, Abd-Elmegid Laila A, Mahmoud Hamdi A, Haggag Mohamed H. Optimized k-means clustering model based on gap statistic. *Int J Adv Comput Sci Appl* 2019;10(1). <http://dx.doi.org/10.14569/IJACSA.2019.0100124>.
- [62] Mohajer Mojan, Englmeier Karl-Hans, Schmid Volker J. A comparison of gap statistic definitions with and without logarithm function. 2011, arXiv preprint arXiv:1103.4767.
- [63] CaliNski Tadeusz. *Dendrogram*. Wiley StatsRef: Statistics Reference Online; 2014.

- [64] Langfelder Peter, Zhang Bin, Horvath Steve. Defining clusters from a hierarchical cluster tree: the dynamic tree cut package for R. *Bioinformatics* 2008;24(5):719–20.
- [65] Nielsen Frank, Nielsen Frank. Hierarchical clustering. In: *Introduction to HPC with MPI for data science*. 2016, p. 195–211. http://dx.doi.org/10.1007/978-3-319-21903-5_8.
- [66] Li Hui, Brouwer Cory R, Luo Weijun. A universal deep neural network for in-depth cleaning of single-cell rna-seq data. *Nat Commun* 2022;13(1):1901. <http://dx.doi.org/10.1101/2020.12.04.412247>.
- [67] Asante-Mensah Maame G, Phan Anh Huy, Ahmadi-Asl Salman, Aghbari Zaher Al, Cichocki Andrzej. Image reconstruction using superpixel clustering and tensor completion. 2023.
- [68] Dunn Joseph C. Well-separated clusters and optimal fuzzy partitions. *J Cybern* 1974;4(1):95–104.
- [69] Karim Md Rezaul, Beyan Oya, Zappa Achille, Costa Ivan G, Rebholz-Schuhmann Dietrich, Cochez Michael, Decker Stefan. Deep learning-based clustering approaches for bioinformatics. *Brief Bioinform* 2021;22(1):393–415. <http://dx.doi.org/10.1093/bib/bbz170>.
- [70] Higham Desmond J, Kalna Gabriela, Kibble Milla. Spectral clustering and its use in bioinformatics. *J Comput Appl Math* 2007;204(1):25–37. <http://dx.doi.org/10.1016/j.cam.2006.04.026>.
- [71] Olman Victor, Mao Fenglou, Wu Hongwei, Xu Ying. Parallel clustering algorithm for large data sets with applications in bioinformatics. *IEEE/ACM Trans Comput Biol Bioinform* 2008;6(2):344–52. <http://dx.doi.org/10.1109/TCBB.2007.70272>.
- [72] Haraty Ramzi A, Dimishkieh Mohamad, Masud Mehedi. An enhanced k-means clustering algorithm for pattern discovery in healthcare data. *Int J Distrib Sens Networks* 2015;11(6):615740. <http://dx.doi.org/10.1155/2015/6157>.
- [73] Delias Pavlos, Doumpos Michael, Grigoroudis Evangelos, Manolitzas Panagiotis, Matsatsinis Nikolaos. Supporting healthcare management decisions via robust clustering of event logs. *Knowl-Based Syst* 2015;84:203–13. <http://dx.doi.org/10.1016/j.knsys.2015.04.012>.
- [74] Yang Shuiqiao, Huang Guangyan, Cai Borui. Discovering topic representative terms for short text clustering. *IEEE Access* 2019;7:92037–47.
- [75] Yin Hui, Song Xiangyu, Yang Shuiqiao, Huang Guangyan, Li Jianxin. Representation learning for short text clustering. In: *Web information systems engineering-WISE 2021: 22nd international conference on web information systems engineering, WISE 2021, Melbourne, VIC, Australia, October 26–29, 2021, proceedings, part II 22*. Springer; 2021, p. 321–35.
- [76] Cao L, Zhao Z, Wang D. Clustering algorithms chapter 5. Singapore: Springer; 2023. http://dx.doi.org/10.1007/978-981-99-1533-0_5.
- [77] Dhanachandra Nameirakpam, Manglem Khumanthem, Chanu Yambem Jina. Image segmentation using k-means clustering algorithm and subtractive clustering algorithm. *Procedia Comput Sci* 2015;54:764–71. <http://dx.doi.org/10.1016/j.procs.2015.06.090>.
- [78] Jose Teena, Babu Suvanam Sasidhar. Detecting spammers on social network through clustering technique. *J Ambient Intell Humaniz Comput* 2019;1–15. <http://dx.doi.org/10.1007/s12652-019-01541-6>.
- [79] Zhao Peixin, Zhang Cun-Quan. A new clustering method and its application in social networks. *Pattern Recognit Lett* 2011;32(15):2109–18. <http://dx.doi.org/10.1016/j.patrec.2011.06.008>.
- [80] Li Pan, Dau Hoang, Puleo Gregory, Milenkovic Olga. Motif clustering and overlapping clustering for social network analysis. In: *IEEE INFOCOM 2017 - IEEE conference on computer communications*. 2017, p. 1–9. <http://dx.doi.org/10.1109/INFOCOM.2017.8056956>.
- [81] Alom Md Zahangir, Tarek M Taha. Network intrusion detection for cyber security using unsupervised deep learning approaches. In: *2017 IEEE national aerospace and electronics conference. NAECON, 2017*, p. 63–9. <http://dx.doi.org/10.1109/NAECON.2017.8268746>.
- [82] Das Rishabh, Thomas H Morris. Machine learning and cyber security. In: *2017 international conference on computer, electrical & communication engineering. ICCECE, 2017*, p. 1–7. <http://dx.doi.org/10.1109/ICCECE.2017.8526232>.
- [83] Kolini Farzan, Janczewski Lech. Clustering and topic modelling: A new approach for analysis of national cyber security strategies. 2017.
- [84] Landauer Max, Skopik Florian, Wurzenberger Markus, Rauber Andreas. System log clustering approaches for cyber security applications: A survey. *Comput Secur* 2020;92:101739. <http://dx.doi.org/10.1016/j.cose.2020.101739>.
- [85] Syarif I, Prugel-Bennett A, Wills G. Unsupervised clustering approach for network anomaly detection. In: *Networked digital technologies. Communications in computer and information science*, vol. 293, Berlin, Heidelberg: Springer; 2012. http://dx.doi.org/10.1007/978-3-642-30507-8_13.
- [86] Markovitz Amir, Sharir Gilad, Friedman Itamar, Zelnik-Manor Lihi, Avidan Shai. Graph embedded pose clustering for anomaly detection. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, p. 10539–47.
- [87] Aggarwal Charu C, Zhai ChengXiang. A survey of text clustering algorithms. *Min Text Data* 2012;77–128. http://dx.doi.org/10.1007/978-1-4614-3223-4_4.
- [88] Aljalbout Elie, Golkov Vladimir, Siddiqui Yawar, Strobel Maximilian, Cremers Daniel. Clustering with deep learning: Taxonomy and new methods. 2018. <http://dx.doi.org/10.48550/arXiv.1801.01587>, arXiv preprint [arXiv:1801.07648](https://arxiv.org/abs/1801.07648).
- [89] Shaham Uri, Stanton Kelly, Li Henry, Nadler Boaz, Basri Ronen, Kluger Yuval. Spectralnet: spectral clustering using deep neural networks. 2018. <http://dx.doi.org/10.48550/arXiv.1801.01587>, arXiv preprint [arXiv:1801.01587](https://arxiv.org/abs/1801.01587).
- [90] Bianchi Filippo Maria, Grattarola Daniele, Alippi Cesare. Spectral clustering with graph neural networks for graph pooling. In: Hal Daumé III, Singh Aarti, editors. *Proceedings of the 37th international conference on machine learning. Proceedings of machine learning research*, vol. 119, Cambridge, MA, USA: PMLR; 2020, p. 874–83, URL <https://proceedings.mlr.press/v119/bianchi20a.html>.
- [91] Sinaga Kristina P, Yang Miin-Shen. Unsupervised k-means clustering algorithm. *IEEE Access* 2020;8:80716–27. <http://dx.doi.org/10.1109/ACCESS.2020.2988796>.
- [92] Yu Shyr-Shen, Chu Shao-Wei, Wang Chuin-Mu, Chan Yung-Kuan, Chang Ting-Cheng. Two improved k-means algorithms. *Appl Soft Comput* 2018;68:747–55. <http://dx.doi.org/10.1016/j.asoc.2017.08.032>.
- [93] Fard Maziar Moradi, Thonet Thibaut, Gaussier Eric. Deep k-means: Jointly clustering with k-means and learning representations. *Pattern Recognit Lett* 2020;138:185–92. <http://dx.doi.org/10.1016/j.patrec.2020.07.028>.
- [94] Ran Xiaojuan, Zhou Xiangbing, Lei Mu, Tepsan Worawit, Deng Wu. A novel k-means clustering algorithm with a noise algorithm for capturing urban hotspots. *Appl Sci* 2021;11(23):11202. <http://dx.doi.org/10.3390/app112311202>.